INCH-POUND

A-A-54928
30 November 1993

COMMERCIAL ITEM DESCRIPTION

ORTHOPEDIC SYSTEM, SURGICAL, ELECTRIC POWER

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers an electric powered, surgical, orthopedic system for performing large bone orthopedic procedures.

Salient characteristics:

The system shall be an electric powered orthopedic system suitable for performing large bone orthopedic procedures. It shall consist of the following components:

* Electric powered orthopedic instrumentation
* Electric motor
* Motor housing
* Holder for flexible cable
* Mobile stator with stainless steel mesh basket
* Footswitch with wall socket cord
* Sterilization cases, filters, locks and cords
* Instrument lubricant
* Packing container with racks and inserts

The orthopedic instrumentation shall consist of the following items and their accessories:

* One each drill with electric cable, torque handle, one 1/4 inch Jacobs chuck with key and one Phillips head screwdriver, for drilling, driving taps, screws and wires.

* One each reamer with Hudson adapter, for use in intramedullary reaming procedures.

AMSC N/A FSC 6515

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* One each reciprocating saw, with three different size reciprocating saw blades, for performing transverse bone procedures.

* One each oscillating saw, with one Allen wrench and five different size oscillating saw blades, for longitudinal bone procedures.

* Two each wire drivers, with chuck (if necessary) for insertion of Steinmann pins and Kirschner wires.

The drill shall be powered by electricity and shall supply sufficient power for most orthopedic drilling, wire driving and pinning procedures. It shall be small and lightweight in design, facilitating use, maneuverability, and surgeon comfort. It shall weigh not more than 2-1/2 pounds. The exterior shall have a satin finish to reduce glare. No-load speed shall be approximately 0-900 RPM in both forward and reverse position. The motor shall be activated by a finger control trigger located at the inward base of the handle. It shall have variable speed with directional control for drill, screw and reverse positions that permits one-handed operation as well as simple shift from forward to reverse. It shall have a safety lock feature to prevent accidental engagement of the trigger when not in use. The drill shall have a snap-lock head. It shall include two packages (6 each) of stainless steel twist drill bits, 7/64 inch in diameter with round distal ends, to fit a Jacobs chuck.

The drill shall have a quick release Jacobs chuck fully cannulated to accept drills, pins, and wires from 1/32 up to 1/4 inch in diameter. It shall have a quick release chuck for items with a square shank (i.e., drills, taps, and screw drivers). It shall have a quick release chuck for items with an AO shank (i.e., drills, taps, and screw drivers). It shall have a quick release chuck for Kirschner wires up to 2.5 mm in diameter.

The drill shall have one each of the following, compatible with the quick release adapters:

* Small Phillip's head screwdriver
* Large Phillip's head screwdriver
* Screwdriver for screws with hex recessed head, 2.5 mm
* Screwdriver for screws with hex recessed head, 3.5 mm

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The system shall be supplied with five stainless steel sterilization containers, each designed to hold, sterilize and store one instrument and its accessories. Each rigid container shall have an inner basket for sterilization and storage of the instrumentation. The container shall be perforated on both the top and bottom to allow sterilization in a gravity displacement sterilizer. The container shall have accommodations for a filter at all perforation sites. The inner basket shall be fabricated of a heavy duty stainless steel wire. The containers shall be approximately 12 inches wide, 10 inches deep, and 4 inches high, shall be flat surfaced to facilitate stacking and storage, and shall have a hinged top with a simple flip lock closure mechanism for securing the lid during handling and sterilization. The lock shall be located on the front panel.

The electric motor shall operate from 110/220 volt, 50/60 Hz power.

The motor housing shall be fabricated of a sterilizable metal material. The sterile housing shall hinge-open to fit over the motor to produce a sterile field around the motor. The holder for the cable shall fit into the motor through the housing.

It shall have a lightweight, flexible, detachable electric cable which locks into the base of the instrument handle and serves as the conduit for the power. The cord shall be at least 6 feet in length and shall connect to the power source on the distal end.

The reamer shall be powered by electricity and shall supply sufficient power to perform intramedullary reaming. It shall be small and lightweight in design, facilitating use, maneuverability and surgeon comfort. It shall weigh not more than 2-1/2 pounds. The exterior shall have a satin finish to reduce glare. The no-load speed shall be approximately 150-300 RPM in both forward and reverse position. The reamer shall be compatible with the power cord described above.

The reciprocating saw shall be electrically powered and shall provide sufficient power to perform transverse bone cutting procedures. It shall be small and lightweight in design, facilitating use, maneuverability, and surgeon comfort. It shall weigh not more than 2 pounds. The exterior of the instrument shall have a satin finish to reduce glare. Speed shall be approximately 18,000 cyles/minute. The blades shall

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attach with or without tools and shall be capable of being set in any position of a 360 degree rotation. It shall have the following reciprocating blades.

* Heavy duty (3 each)
* Regular use (3 each)
* Short length (3 each)

If a tool is needed to attach the blades, it shall be compatible with the saw and come with the instrumentation. The saw shall be compatible with the power cord described above.

The oscillating saw shall be electrically powered and shall supply sufficient power to perform long bone procedures. It shall be small and lightweight in design, facilitating use, maneuverability and surgeon comfort. It shall weigh not more than 2-1/2 pounds. The exterior of the instrument shall have a satin finish to reduce glare. No-load speed shall be approximately 18,000 cycles/minute. The saw blades shall lock to the head by use of a wrench. The head of the instrument shall rotate 360 degrees with at least 12 different settings. The blade shall be capable of rotating at least 180 degrees. The following size oscillating blades shall be included with the saw. The blades shall be fabricated of stainless steel and shall have 18 to 20 teeth per inch.

* Straight blade 62 mm long (3 each)
* Straight blade 75 mm long (3 each)

The saw shall be compatible with the power cord described above.

The mobile stator shall consist of a mobile base with a longitudinal tube onto which the wire basket and the motor can be attached. The woven stainless steel mesh basket shall clamp onto the mobile stator to hold the sterile instrumentation during use.

The system shall include filters, locks and cords necessary for 1000 sterilization processes using the rigid containers.

The entire system, including the instrumentation, the motor, the stator, and all cords and cables shall be housed in a packing container with racks and inserts, which shall store and protect all items against damage. The container shall be

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fabricated of a high strength sheet aluminum alloy which is resistant to temperatures of minus 40°C to minus 70°C. It shall have spring-loaded aluminum handles and a rubber seal on top. The weight of the system shall be approximately 26 pounds.

The system shall contain one can of Kavo, or equal, spray for lubrication of instruments.

All metal instrumentation shall be fabricated of stainless steel. All other metal parts shall be fabricated of a suitable corrosion-resistant metal alloy that has been properly hardened and passivated.

The system shall be furnished with at least two copies of instructions for operation, maintenance and repair.

The unit shall meet Underwriters Laboratory Code 544 Medical and Dental Equipment (UL544) and shall be approved by Underwriters Laboratory or an approved Government testing laboratory.

Workmanship. The electric power, surgical, orthopedic system shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One electric power, surgical, orthopedic system, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 0.0 (percent defective) and an inspection level of S-2. The unit of product for examination purposes shall be one electric power, surgical, orthopedic system.

Sampling for tests. Sampling for tests, including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 0.0 (percent defective) and an inspection level of S-2. The unit of product for test purposes shall be one electric power, surgical, orthopedic system.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics. Tests shall include, but shall not be limited to, the following:

* Test drill speed for 0 to 900 RPM range, forward and reverse.
* Test reamer speed for 150 to 300 RPM range, forward and reverse.
* Test reciprocating saw speed for 18,000 cycles/minute.
* Test oscillating saw speed for 18,000 cycles/minute.
* Test the electric motor operation from 110/220 VAC, 50/60 Hz with plus or minus 10 percent VAC, plus or minus 2 Hz.
* Test the unit per UL 544 requirement for safety for use in an operating room (i.e. dielectric test, grounding, leakage, etc.).

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

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The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers a and/or contents thereof under normal shipping conditions, handling, etc , involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

6515-01-323-2647

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MILITARY INTERESTS:

Custodians:

Air Force - 03
Army - MD
Navy - MS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

FDA-MPQAS
USPHS
VA - OSS

Project Number: 6515-5625

Location: ORTHO17C.DOC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I. RECOMMEND A CHANGE:	1. DOCUMENT NUMBER: A-A-54928	2. DOCUMENT DATE (YYMMDD): 30 November 1993

3. DOCUMENT TITLE: ORTHOPEDIC SYSTEM, SURGICAL, ELECTRIC POWER

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

7. PREPARING ACTIVITY

a. NAME DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-MSE	b. TELEPHONE (Include Area Code) (1) Commercial: AC 215-737-2870	(2) AUTOVON: AV 444-2870
c. ADDRESS (Include Zip Code) 2800 SO. 20TH STREET PHILADELPHIA, PA 19145	8. IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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